P3-005

Referent-Configuration Imitation Learning (RCIL) for Predictive Musculoskeletal Simulation

Faster and more stable gait imitation by controlling referent configurations instead of muscle activations

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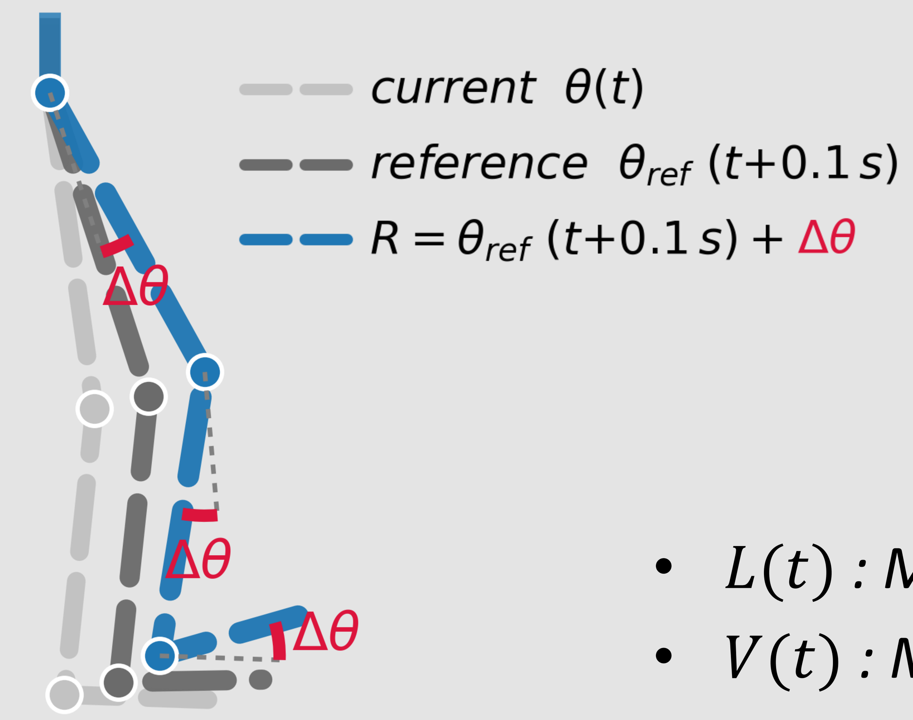
PROBLEM

- Predictive musculoskeletal simulation (MSK) is powerful, but muscle redundancy makes control hard [1].
- Imitation learning reproduces motion [2], yet controlling in muscle-activation space is slow, unstable, and prone to collapses [3].

MOTIVATION

- The CNS controls movement by shifting referent configurations (λ-thresholds), rather than muscle activations, a low-dimensional and self-stabilizing control variable [4, 5].
  - **Our idea: let the policy learn in referent configuration space — a small residual Δθ added to the reference pose.**
- The 0.1 s anticipatory lead matches the ~100 ms neuromechanical delay that the CNS compensates for [6].

METHODS & RESULTS



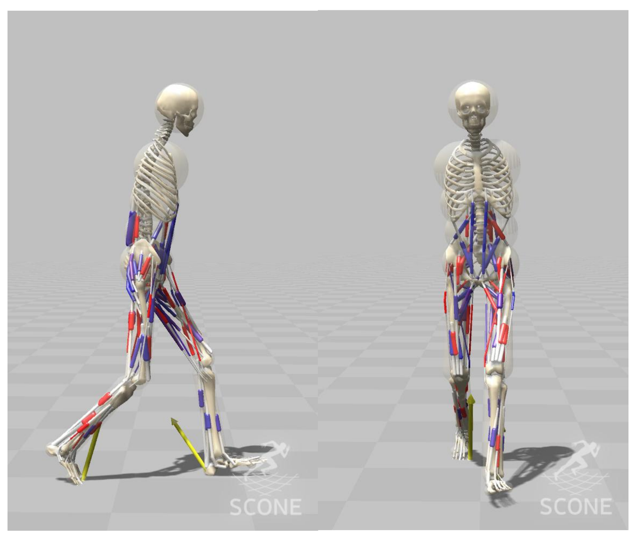
1) $R = \theta_{ref}(t + 0.1s) + \Delta\theta$ ➢ R is the target referent configuration.

2) λ_R = *muscle fiber length* (R) ➢ λ_R is the muscle fiber length at R.

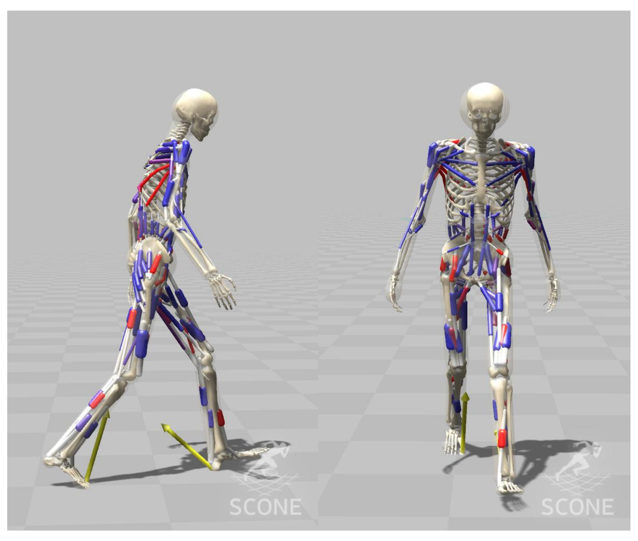
3) $A(t) = [g(L(t) - (\lambda_R - \delta) + \mu V(t)]_0^1$ ➢ $A(t)$ is the muscle activation [4, 5].

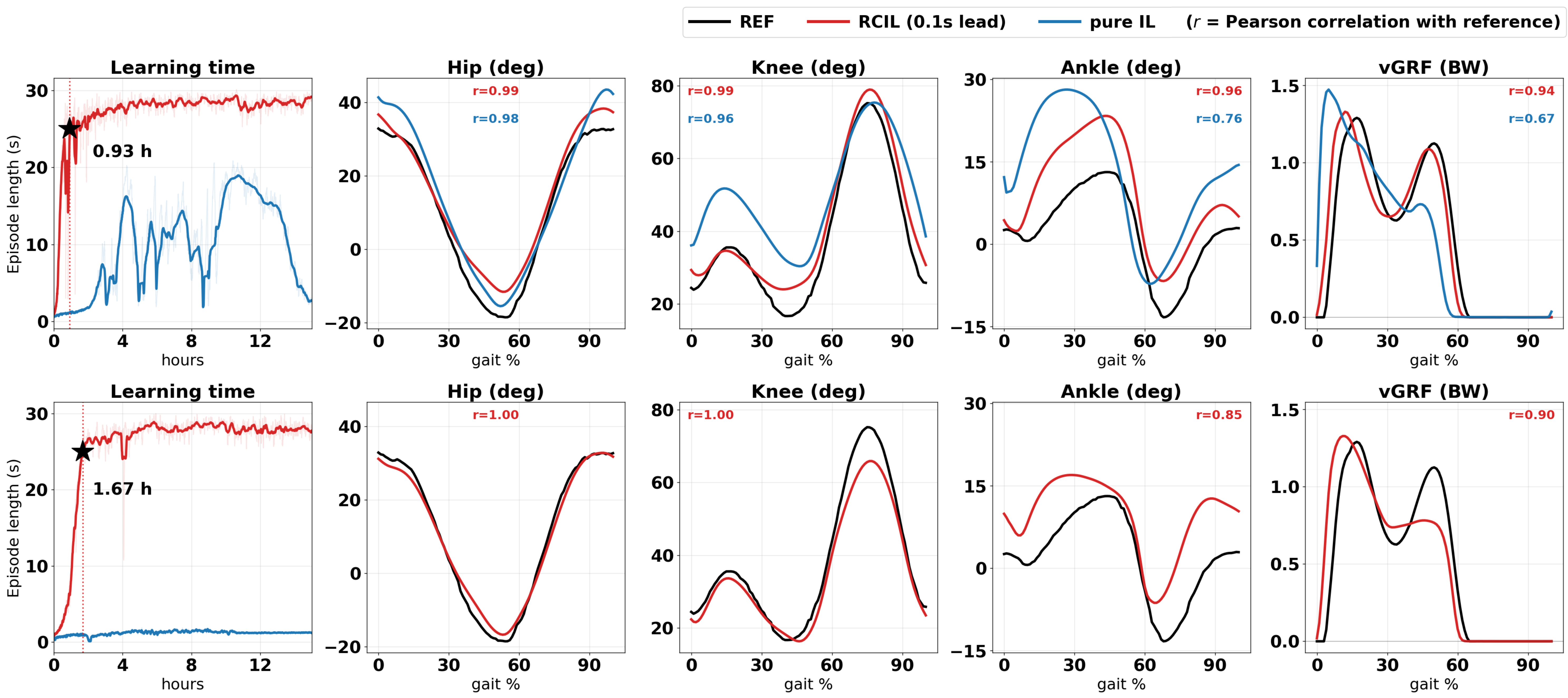
- $L(t)$: *Muscle fiber length*
- $V(t)$: *Muscle fiber Velocity*
- $[g]_0^{20}$: *Length-based reflex (tonic) gain*
- $[\mu]_{-0.1}^{+0.1}$: *Velocity-based reflex (phasic) gain*
- $[\delta]_0^{0.2}$: *Baseline muscle tone*
- $(\lambda_R - \delta)$: *Length-threshold*

H2190 (SCONE)



H32130 (SCONE)





Check this out!

- **RCIL reached stable walking within ~1-2 hours in both models** (H2190 0.93 h, H32130 1.67 h), whereas pure IL catastrophically forgot (H2190) or never converged (H32130).
- **RCIL tracked reference joint kinematics closely across both models** (hip & knee r ≥ 0.99, ankle r = 0.96/0.85), while pure IL degraded distally (H2190 ankle r = 0.76).
- **RCIL reproduced the vertical GRF** (r = 0.94/0.90), far above pure IL, which lost push-off (H2190 vGRF r = 0.67).

DISCUSSION & CONCLUSION

➢ Commanding referent configurations instead of muscle activations turns muscle redundancy into a low-dimensional, dynamically consistent target, so RCIL converges quickly to stable walking while direct imitation does not.

➢ RCIL turns motion data into subject-specific musculoskeletal digital twins that are fast to train, accurate, and ready for predictive simulation.

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REFERENCES

[1] Geijtenbeek et al. (2019), SCONE, [2] Peng et al. (2018) [3] Song et al. (2021), [4] Feldman (1986), [5] St-Onge (2004) [6] Afschrift et al. (2021)